BEWLEY-AIYAGARI-HUGGETT-IMROHOROGLU MODELS

Quantitative Economics 2025

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INTRODUCTION

MOTIVATION

- We introduce a class of models with heterogeneous agents.
- We focus on models with household heterogeneity and incomplete markets: Aiyagari-Bewley-Huggett-Imrohoroglu models.
  - Infinitely lived households face idiosyncratic shocks
  - Markets are incomplete – households cannot trade all assets they would like to trade
- We will find general equilibrium of these models.

HOUSEHOLD PROBLEM

AGENTS

- In all these models there is a continuum of agents/households indexed by $i \in [0, 1]$.
- Agents are infinitely lived, have a standard period utility function $u(c)$, where c is consumption, discount future at $\beta \in (0, 1)$.
- Labor endowment (productivity) is stochastic and follows a stationary Markov process (same process for all i), with a finite set of states Z and transition matrix $Q(z, z')$.
- Agents consume and trade a single asset a that pays (net) return r. Wage per unit of labor is w.
- Agents face a borrowing constraint, $a \geq -\phi$.

HOUSEHOLD PROBLEM

- **State variables:** beginning of period assets a and labor endowment z.
- Recursive formulation of a problem of an agent with assets a and labor endowment z is

$$
\begin{aligned}
v(a,z) = \max_{c,a'} \left\{ u(c) + \beta \sum_{z' \in Z} Q(z,z') v(a',z') \right\} \\
\text{subject to } c + a' = zw + (1+r)a, \\
a' \geq -\phi.
\end{aligned}
$$

- The solution consists of the value function $v(a,z)$ and the policy functions $a'(a,z)$, $c(a,z)$.
- **Note:** Everything (e.g. v) depends on (r,w), we suppress it in notation.

DISTRIBUTION

- Suppose that there is a finite grid of asset levels $\{a_1, \ldots, a_N\}$ and define the unconditional distribution $\lambda_t(a, z) := \mathbb{P}(a_t = a, z_t = z)$.
- Given the policy function $a'(a, z)$ and Q we have

$$\lambda_{t+1}(a', z') = \sum_{a,z} \lambda_t(a, z) \cdot Q(z, z') \cdot \mathbb{I}(a' = a'(a, z)),$$

where $\mathbb{I}$ is the indicator function.
- What if assets are not restricted to a grid? Similar logic, but a more messy formula (+ measure theory).
- We will need to discretize the state space and action space anyway.

AGGREGATION

- Given a distribution λ_t and policy functions $a'(a,z)$ and $c(a,z)$ we can calculate aggregate variables.
- We have asset demand and consumption

$$\begin{aligned} A'_t &= \int a'(a,z) \cdot d\lambda_t(a,z), \\ C_t &= \int c(a,z) \cdot d\lambda_t(a,z). \end{aligned}$$

- For a finite grid we have $A'_t = \sum_{a,z} a'(a,z) \cdot \lambda_t(a,z)$ and $C_t = \sum_{a,z} c(a,z) \cdot \lambda_t(a,z)$.

HUGGETT MODEL

THE REST OF THE ECONOMY

- How are r and w determined?
- What/who supplies goods?
- What/who supplies assets?
- We will start with the Huggett model:
  - Endowment economy: w exogenous, wz is the amount of goods received by an agent
  - Nothing/nobody else in the economy. Supply of assets is zero
- Market clearing conditions:
  - Goods market: $C_t = w \int z \, d\lambda_t(a, z)$
  - Asset market: $A_t' = 0$

HUGGETT MODEL

- Assets are loans from/to agents.
- Asset market clears if the total amount of loans (negative a) is equal to the total amount of savings (positive a).
- We will be looking for a stationary equilibrium: the distribution of agents and prices (here only r) are constant over time.
- This was already anticipated by how we wrote the Bellman equation – no time subscripts anywhere.

RCE IN HUGGETT MODEL

Definition (Stationary Recursive Competitive Equilibrium (RCE))

A stationary recursive competitive equilibrium is a rate of return r, a value function $v(a, z)$, policy functions $a'(a, z)$ and $c(a, z)$, and a distribution $\lambda(a, z)$ such that

1. Given r, the value function $v(a, z)$ satisfies the Bellman equation, and associated policy functions $a'(a, z)$, $c(a, z)$, solve the agent's maximization problem;
2. The probability distribution $\lambda(a, z)$ is the stationary distribution of the Markov process (a_t, z_t) induced by Q and $a'(a, z)$;
3. Markets clear:

$$\int a'(a, z) \cdot d\lambda(a, z) = 0, \quad \int c(a, z) \cdot d\lambda(a, z) = w \int z \, d\lambda(a, z).$$

COMMENTS

- We now need to find r such that aggregate asset demand resulting from optimal decisions of agents is zero.
- Alternatively, we can find r such that the goods market clears.
- Recall: the Walras law says that if there are N markets and $N-1$ clear, the N-th market also clears.

COMPUTATIONAL APPROACH

ALGORITHM FOR FINDING EQUILIBRIUM

- The usual procedure is:
  1. Guess r.
  2. Solve the Bellman equation for $v(a,z)$ and $a'(a,z)$.
  3. Find the stationary distribution $\lambda(a,z)$.
  4. Calculate aggregate asset demand A'.
  5. If $A' > 0$, decrease r; if $A' < 0$, increase r.
- This combines several things:
  - solving the Bellman equation
  - finding the stationary distribution
  - finding the equilibrium price r

COMPUTATION

- When we solve the model on a computer we discretize the state space and have a finite grid of points for assets: $\{a_1, \ldots, a_N\}$ and for productivity $\{z_1, \ldots, z_M\}$.
- We usually want to allow maximizers of the RHS of the Bellman equation to not necessarily belong to the grid.
- We cannot use the formula

$$\lambda(a', z') = \sum_{a,z} \lambda(a, z) \cdot Q(z, z') \cdot \mathbb{I}(a' = a'(a, z)),$$

  because $a'(a, z)$ might not belong to the grid.
- How to find the stationary distribution $\lambda(a, z)$ in this case?

YOUNG (2010) METHOD

YOUNG (2010): OFF-GRID POLICY FUNCTIONS

- Let

$$q(a,z,a_n) = \mathbb{I}(a'(a,z) \in [a_{n-1},a_n])\frac{a'(a,z)-a_{n-1}}{a_n-a_{n-1}} + \mathbb{I}(a'(a,z) \in [a_n,a_{n+1}])\frac{a_{n+1}-a'(a,z)}{a_{n+1}-a_n}$$

  be the weight assigned to grid point a_n.
- Then

$$\lambda(a',z') = \sum_{a,z} \lambda(a,z) \cdot Q(z,z') \cdot q(a,z,a').$$

- This is the same as saying that the fraction $q(a,z,a_n)$ of agents with assets a and productivity z will choose a_n.
- This approach (due to Young (2010)) is useful because it makes aggregates unbiased.

AGGREGATION AND EQUILIBRIUM

- Once we have the distribution $\lambda(a, z)$ we can calculate aggregates.
- For example:

$$A' = \sum_{a,z} a'(a, z) \cdot \lambda(a, z).$$

- We can repeat the same procedure for various r to find the equilibrium r, such that $A' = 0$.
- Better: write a function that takes r as an input and returns A' as an output. Then use a root finding algorithm to find the equilibrium r.

DISCUSSION: BOUNDS ON INTEREST RATE

- Suppose we want to use a bracketing method to find the equilibrium r.
- What are the appropriate bounds? Probably $r < -1$ does not make sense (nobody would save). The upper bound is more tricky.
- We can show that $r < \beta^{-1} - 1$ in equilibrium for the Huggett model (just see what happens if $r > \beta^{-1} - 1$).
- **Intuition:** If $r > \beta^{-1} - 1$, then $(1 + r)\beta > 1$, so agents would want to save infinitely, which cannot be an equilibrium.

DISCUSSION: COMPUTATIONAL TIPS

- Here you need to solve the Bellman equation for each r: possibly many times.
- This can be costly - try to optimize the code.
  - Use HPI or OPI (Howard Policy Iteration or Optimistic Policy Iteration)
  - Use EGM (Endogenous Grid Method)
  - Get the transition matrix for (a, z) and use it to find the stationary distribution (do not simulate anything!)
  - Check out Simon Mongey's slides – general notation but similar models

AIYAGARI MODEL

AIYAGARI MODEL: PRODUCTION

- In Aiyagari model there is a representative firm that hires labor and rents capital from households.
- The firm has a constant returns to scale production function $F(K, L)$, where K is capital and L is labor.
- The firm is competitive, so it takes r and w as given.
- The firm's problem is

$$\max_{K,L} F(K, L) - (r + \delta)K - wL,$$

where δ is the depreciation rate.

AIYAGARI MODEL: MARKET CLEARING

- Assets accumulated by households are capital and loans to other agents.
- Labor market clearing: $L = \int z \, d\lambda(a,z)$.
- Asset market clearing: $K = \int a \, d\lambda(a,z)$.
- Goods market clearing: $F(K,L) = C + \delta K$.

AIYAGARI MODEL: PRICES

- Notice that here we have

$$r = F_K(K,L) - \delta, \quad w = F_L(K,L).$$

- Because of the constant returns to scale

$$r = F_K\left(\frac{K}{L},1\right) - \delta, \quad w = F_L\left(\frac{K}{L},1\right).$$

- We can solve for K/L as a function of r. This also allows us to solve for w as a function of r.
- We know L (it is exogenous) so we have $K(r)$.

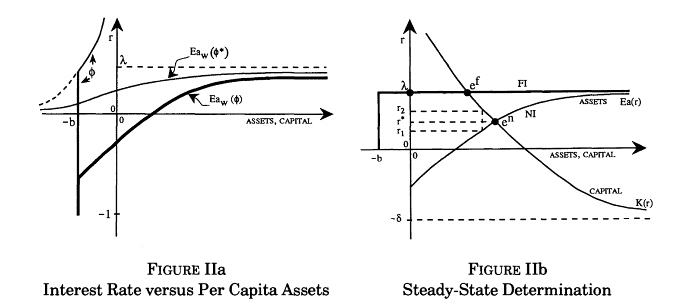
AIYAGARI MODEL: ALGORITHM

- This suggests the following strategy:
    1. Guess r. Repeat all the steps from the Huggett model to find A'.
    2. Calculate $K(r)$ from the firm's problem.
    3. Check if $A' = K$. If yes, we found the equilibrium r. If not, adjust r and repeat.
- We can simply find the root of $A'(r) - K(r) = 0$.

AIYAGARI MODEL: KEY RESULTS

- **Precautionary savings:** With incomplete markets and idiosyncratic risk, agents save more than in a complete markets economy.
- This leads to:
  - Higher capital stock K than in complete markets
  - Lower interest rate r than in complete markets
  - Higher wages w (capital and labor are complements)
- **Intuition:** Agents cannot insure against idiosyncratic risk, so they accumulate assets as a buffer against bad shocks.
- The equilibrium interest rate is below the rate of time preference: $r < \beta^{-1} - 1$.

AIYAGARI MODEL: EQUILIBRIUM DETERMINATION



Equilibrium determination in Aiyagari model (Aiyagari, 1994, QJE)

- Left panel: $Ea_w(\phi)$ is aggregate asset demand as function of borrowing constraint
- Right panel: Intersection of asset demand $Ea(r)$ and capital demand $K(r)$ determines equilibrium

AIYAGARI MODEL: EFFECTS OF BORROWING CONSTRAINTS

- The borrowing constraint ϕ affects the equilibrium:
  - Tighter constraint (lower ϕ) $\Rightarrow$ less borrowing $\Rightarrow$ higher asset demand
  - Higher asset demand $\Rightarrow$ higher K $\Rightarrow$ lower r
- With very tight constraints, some agents are liquidity constrained.
- The model generates:
  - Realistic wealth distribution (right-skewed)
  - Positive correlation between wealth and labor income
  - Precautionary savings behavior

ADDING GOVERNMENT

ADDING GOVERNMENT: MOTIVATION

- In many applications we are interested in the effects of some government policies.
- Example: how does an increase in taxation affect the wealth distribution?
- Example: how does an increase in government debt crowd out capital accumulation?
- We will now consider a simple extension of the Aiyagari model with government.

GOVERNMENT BUDGET CONSTRAINT

- The intertemporal budget constraint of the government is

$$B_{t+1} = (1+r)B_t + T_t - G_t,$$

  where B_t is the government debt, T_t is tax revenue net of transfers and G_t is government purchases of goods.
- The government collects taxes on labor and capital income. Linear tax system with rates τ^w, τ^r. Tax revenue net of transfers is

$$T = \tau^w w \int z \, d\lambda(a,z) + \tau^r r \int a \, d\lambda(a,z) - d,$$

  where d is lump-sum transfers.

GOVERNMENT: STATIONARY EQUILIBRIUM

- In a stationary equilibrium the government budget constraint becomes

$$rB = T - G.$$

- Asset market clearing: $K + B = \int a \, d\lambda(a, z)$.
- Goods market clearing: $F(K, L) = C + \delta K + G$.
- **Key difference:** assets available in the economy are $K + B$, not just K.

HOUSEHOLD PROBLEM WITH TAXATION

- We also need to modify the household problem.
- Recursive formulation of a problem of an agent with assets a and labor endowment z is

$$v(a,z) = \max_{c,a'} \left\{ u(c) + \beta \sum_{z' \in Z} Q(z,z') v(a',z') \right\}$$

$$\begin{aligned} \text{subject to } c + a' &= (1-\tau^w) z w + (1 + (1-\tau^r) r) a + d, \\ a' &\geq -\phi. \end{aligned}$$

- **Note:** dependence on government policies, we suppress it in notation.

TRANSITION PATHS

MOTIVATION: TRANSITIONS BETWEEN STEADY STATES

- So far we have studied stationary equilibria where all variables are constant over time.
- In practice, we often want to analyze policy changes or aggregate shocks.
- Example: What happens when the government increases debt from B_0 to B_1?
- Example: What is the transition path after an aggregate TFP shock?
- We need to compute the transition path between two stationary equilibria in the Aiyagari model.

SETUP: PERFECT FORESIGHT TRANSITION

- Economy starts in an initial steady state at $t = 0$ with distribution λ_0.
- At $t = 0$, agents learn about a policy change (e.g., new government debt level B).
- Assume perfect foresight: agents know the entire future path of prices $\{r_t, w_t\}_{t=0}^{\infty}$.
- The economy converges to a new steady state as $t \to \infty$.
- This framework works for:
  - Permanent changes: transition between two different steady states
  - MIT shocks: unexpected one-time shock at $t = 0$, then economy returns to original SS
- Goal: Find the entire transition path $\{K_t, r_t, w_t, \lambda_t\}_{t=0}^{\infty}$ such that markets clear at each t.

KEY CHALLENGE

- In a stationary equilibrium, we solve a fixed point problem: find r such that households' asset supply equals firm's capital demand.
- In a transition, we need to solve a sequence problem: find entire path $\{K_t\}_{t=0}^{T}$ such that markets clear at each t.
- **Problem:** The distribution λ_t depends on the entire past history of prices and policies.
- **Problem:** Households' policy functions $a'(a, z; t)$ depend on the entire future path of prices (perfect foresight).
- This is a high-dimensional fixed point problem.

SIMPLIFICATION: FINITE HORIZON

- Choose a large but finite horizon T such that the economy is "close enough" to the new steady state by period T.
- Assume that for $t \geq T$, the economy is in the new steady state.
- Then we only need to find $\{K_t\}_{t=0}^{T-1}$.
- Check ex-post: Is λ_T close to the new steady state distribution λ^{SS1}? If not, increase T.

WHY GUESS CAPITAL PATH $\{K_t\}$?

- In the Aiyagari model with production function $F(K, L)$, we have firm FOCs:

$$
\begin{aligned}
r_t &= F_K(K_t, L) - \delta, \\
w_t &= F_L(K_t, L).
\end{aligned}
$$

- Labor supply $L = \int z \, d\lambda(a, z)$ is exogenous (depends only on λ, not on prices).
- Therefore: if we know K_t, we can immediately compute (r_t, w_t) from firm FOCs.
- This means we can work with capital path $\{K_t\}$ instead of price paths $\{r_t, w_t\}$.
- Only need to iterate on one sequence instead of two!

ALGORITHM

- **Step 1.** Compute initial steady state (SS0) and final steady state (SS1).
- **Step 2.** Guess a capital path $\{K_t^{(0)}\}_{t=0}^{T-1}$.
  - Simple guess: linear interpolation between K^{SS0} and K^{SS1}
- **Step 3.** Use firm FOCs to compute prices: $(r_t, w_t) = (F_K(K_t, L) - \delta, F_L(K_t, L))$.
- **Step 4.** Solve household problem backwards from $t = T - 1$ to $t = 0$ (details on next slides).
- **Step 5.** Simulate distribution forwards from $t = 0$ to $t = T$.
- **Step 6.** Check market clearing and update capital path if needed.

BACKWARD ITERATION: TIME-DEPENDENT BELLMAN EQUATION

- The household problem at time t with state (a, z) is:

$$v_t(a,z) = \max_{c,a'} \left\{ u(c) + \beta \sum_{z' \in Z} Q(z,z') v_{t+1}(a',z') \right\}$$

$$\text{subject to } c + a' = z w_t + (1 + r_t) a,$$

$$a' \geq -\phi.$$

- Key difference from stationary problem: v_{t+1} is the value function next period, and prices (r_t, w_t) vary with time.
- Solution gives time-dependent policy functions: $a'(a, z; t)$, $c(a, z; t)$.

BACKWARD ITERATION: IMPLEMENTATION

- Start at the terminal period $t = T$:
  - Use final steady state: $v_T(a, z) = v^{SS1}(a, z)$
- For $t = T - 1, T - 2, \ldots, 0$ (backwards):
  - Given $v_{t+1}(a, z)$ and prices (r_t, w_t), solve:

$$v_t(a, z) = \max_{a'} \left\{ u(zw_t + (1 + r_t)a - a') + \beta \sum_{z'} Q(z, z') v_{t+1}(a', z') \right\}$$

  - Store policy function $a'(a, z; t)$ that achieves the maximum
- After backward iteration, we have $\{v_t(a, z), a'(a, z; t)\}_{t=0}^{T-1}$.

FORWARD SIMULATION

- **Step 5.** Obtain distribution forwards from $t = 0$ to $t = T$:
  - Start with $\lambda_0 = \lambda^{SS0}$ (initial steady state distribution)
  - Use policy functions $a'(a, z; t)$ from backward iteration to compute λ_{t+1} from λ_t
- **Step 6.** For each t, compute implied capital supply:

$$\tilde{K}_t = \int a \, d\lambda_t(a, z).$$

- **Step 7.** Check if $\tilde{K}_t \approx K_t$ for all $t = 0, \ldots, T$. If not, update capital path and return to Step 3.

UPDATING THE CAPITAL PATH

- Define excess capital: $EK_t = \tilde{K}_t - K_t$ (implied minus guessed).
- If $EK_t > 0$: households want to save more than guessed $\Rightarrow$ increase K_t.
- If $EK_t < 0$: households want to save less than guessed $\Rightarrow$ decrease K_t.
- Simple updating rule with relaxation parameter $\xi \in (0, 1)$:

$$K_t^{(n+1)} = (1 - \xi)K_t^{(n)} + \xi\tilde{K}_t = K_t^{(n)} + \xi \cdot EK_t$$

- More sophisticated: use adaptive damping or combination of previous guesses, or other root-finding methods.

COMPUTATIONAL TIPS

- Start with a coarse grid and short horizon T to get initial guess, then refine.
- Use adaptive damping: start with large ξ for speed, reduce when close to convergence.
- Check that $\lambda_T \approx \lambda^{SS1}$ and $K_T \approx K^{SS1}$.
- Solving household problem backwards is the most expensive step:
  - Use EGM if possible
  - Parallelize across (a, z) grid points
- **Note:** For MIT shocks, the final steady state equals the initial one (SS1 = SS0), which can simplify computation.